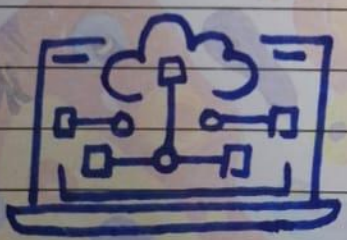
Topic ..... Date : .....

Paper Consumption in the Digital Age

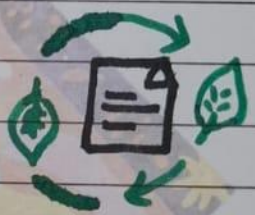
Environmental Impacts and Sustainable Alternative :-



Paper Consumption & Waste



Digital Impact & Emissions



Sustainable Alternatives

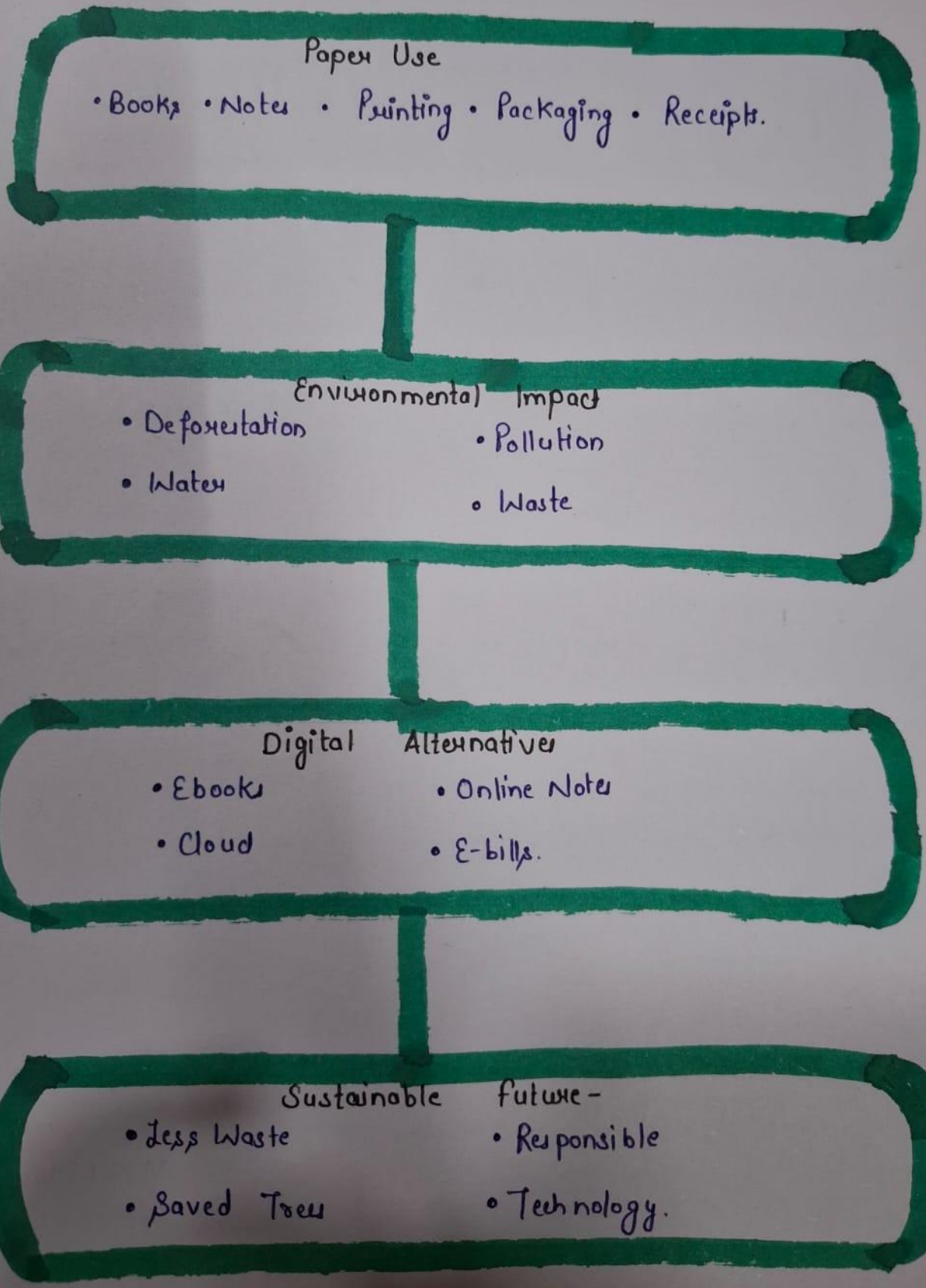
Submitted by -

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Paper Use
• Books • Notes • Printing • Packaging • Receipts.
Environmental Impact
• Deforestation
• Water
• Pollution
• Waste
Digital Alternatives
• Ebooks
• Cloud
• Online Notes
• E-bills.
Sustainable future -
• Less Waste
• Saved Trees
• Responsible
• Technology.

Topic ..... Date : .....

Abstract and Objective

- Paper remains important in education, communication, administration and packaging.
- Digital alternatives such as e-books, cloud storage, online forms and digital payments can reduce unnecessary printing.
- Digital technology also has environmental impacts through electricity use, devices and electronic waste.
- The sustainable choice is to use paper and digital systems responsibly.

- Core Idea - Paper essential but unnecessary printing harms environment. Digital helps but has e-waste cost.

Goal - Responsible use of both.

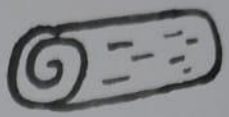
Four Pillar -

1. Paper use - Books, note, printing, packaging
2. Impact - Deforestation, water, pollution, waste.
3. Digital - E-books, cloud, e-bills.
4. Future - Less Waste, saved trees, responsible Tech.

PAPER LIFECYCLE



1. FORESTS



2. WOOD



4. PAPER PRODUCTION



3. PULP



5. Printing



6. USE



7. ReUse • Recycle = • Disposal

Topic ..... Date : .....

Paper Consumption in Daily Life -

- Education - notebooks, textbooks, assignments, worksheets and examination sheets.
- Offices - reports, files, invoices, forms and photocopies.
- Buisnesses - bills, advertisements, brochures and product packaging.
- Unnecessary printing is common when documents could be stored or shared digitally.
- Paper itself is useful; the major concern is unnecessary printing, single-use products and poor recycling.

Paper System

1. RESOURCE USE :

- Uses Trees (raw material from forest)
- Paper produced from wood pulp

2. TRANSPORT

- Requires Physical Transportation.
- Trucks, Cargo Ships & Waterways.
- fuel consumption & emissions during shipping.

3. WASTE OUTPUT

- Generate Physical Paper Waste
- Discarded to Landfill
- Biodegradable but accumulate as Waste.

Digital System

1. RESOURCE USE -

- Uses Electronic Devices
- Metals, Plastics & Rare Minerals
- Requires Manufacturing of Hardware.

2. ENERGY USE -

- Powered by electricity
- Data Centres & Cloud Servers
- Continuous Energy Consumption.

3. WASTE OUTPUT

- Generate E-Waste
- Batteries, Circuits & Components
- Toxic, Hard to recycle

Topic ..... Date : .....

Environmental Impact

(:° Paper Production)

- Paper production requires raw materials, water, energy and chemicals.
- Transport of wood, pulp and finished paper adds energy use and emissions.
- Manufacturing can generate wastewater and other forms of pollution.
- Reducing unnecessary paper demand lowers pressure across the production chain
- High water per ton, energy for drying, pollution if untreated, landfill waste.
- Key Insight - Reducing demand reduces pressure across chain.
- Result - Water waste, air emissions, forest pressure.

Topic ..... Date : .....

Environmental Impact :

(Forests and Biodiversity)

- Trees absorb carbon dioxide, release oxygen, help prevent soil erosion and provide habitats.

Excessive Forest cleaning for wood and pulp can affect biodiversity.

Habitat loss and reduced carbon storage can contribute to climate imbalance.

Impact - Water, Energy and Waste -

Paper production use water and energy at several stages.

Improper disposal increase landfill waste and resource loss.

Recycling helps recover material and reduces the need for fresh resource.

Topic ..... Date : .....

Paper Waste and 5Rs

1. Refuse - Say no to unnecessary printouts and flyers

2. Reduce - Print only essential pages.

3. Reuse - Use one sided sheets for rough work

4. Repurpose - Use old paper and cardboard for projects

5. Recycle - Put clean paper into recycling bins.

Managing paper waste using the 5R's helps protect the environment and cut down on landfill trash.

Topic ..... Date : .....

AI and ML for Sustainable Paper Use

AI can identify documents that can be stored digitally instead of printing.

ML models can predict printing demand and help organisations avoid unneccesary stock.

Digital dashboards can track paper use, waste and recycling performance.

AI can support document classification search and automated workflows.

Shrink large models using pruning, quantization and distillation to use less hardware power.

Run heavy training jobs in data centres that use wind, solar or other green energy sources.

Move computing tasks to hours of the day when clean energy supply is high.

Topic ..... Date : .....

Smart Choice : Paper + Digital

- Keep important or legally required records in appropriate format.
- Prefer digital sharing for collaboration and routine communication.
- Print double-sided and avoid unnecessary copies.
- Recycle clean paper and responsibly dispose of electronic devices.
- Physical classroom provide structured environment while accompanying digital apps allow students to practice vocabulary.
- Storing files on Google Drive, One Drive or Dropbox eliminate the need for physical filling Cabinets and printed layouts.
- Tools like Notion, Slack and Google Doc allow teams to edit documents together without printing multiple drafts.

Topic ..... Date : .....

Sustainable Action Plan

(At LPU)

- Prefer online assignment submission wherever allowed
- Set Printers to double-sided printing by default
- Use QR codes for routine notices, event details and forms.
- Share notes through LMS platform, e-mail or cloud drive.
- Install seperate bins for clean paper collection.
- Use recycled paper for non official printing.
- Encourage students to carry digital copies of notes.
- Run awareness compaign such as "Think Before You Print".
- Track printing use through a digital dashboard.
- Send old electronic devices to authorised e-waste collection system

Topic ..... Date : .....

Conclusion and Bibliography

Paper is useful, but excessive and careless consumption creates environmental problems.

It contributes to pressure on forests, water and energy use, pollution and waste generation.

Digital technology offers useful alternatives for education, administration and communication.

A sustainable future needs both smart technology and responsible behaviour.

Bibliography.

- United Nations Environmental Programme (UNEP)
- Food and Agriculture Organization (FAO)
- Environmental Science textbooks
- Research articles on e-waste and sustainable computer
- College library and educational website.